

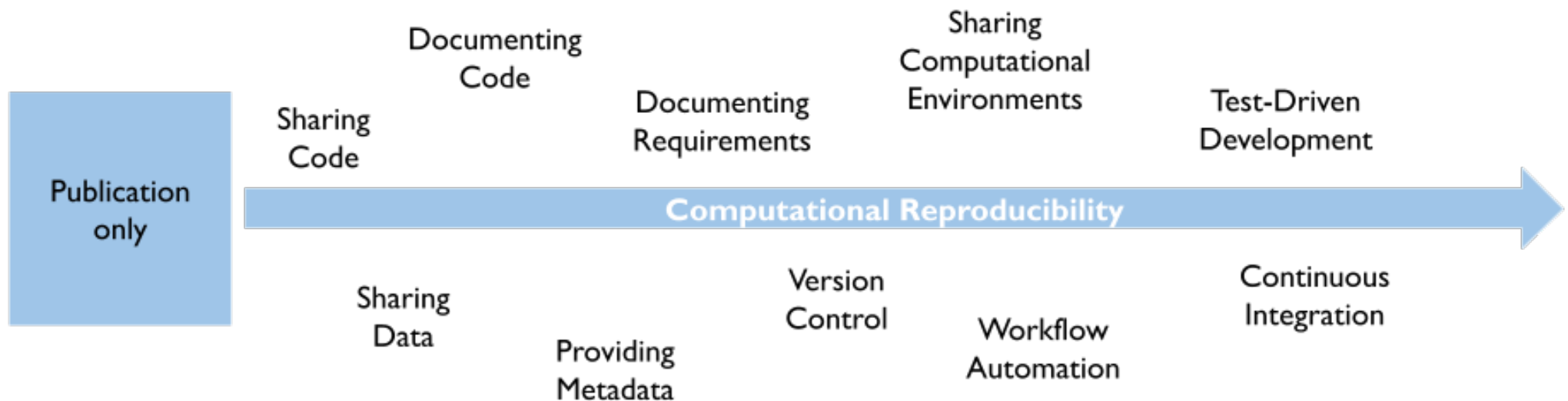
Advanced Workflows

Combining dynamic reporting with version control, automation, and containerization

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Reproducibility is a spectrum



- Goal today: Understanding **what is possible** towards the right side
- ~~Goal today: Deep technical understanding of advanced workflows~~

Dynamic reporting solves copy-paste problem, but ...

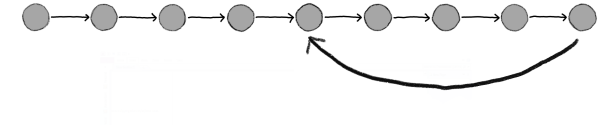
- How to keep track of different **file versions**?
 - `paper-final.qmd` → `paper-final2.qmd` → `paper-final2b.qmd` → 🤯
 - Solution: **Git**
- How to know which files to run in which **order**?
 - `functions.qmd`, `analysis.qmd`, `plots.qmd`, `paper.qmd`, 🤔
 - Solution: **Make**
- How to make sure code still runs when **software versions** change?
 - **Today**: R version 4.5, ggplot2 3.5.2 → **Next year**: R version **?**, ggplot2 **?**
 - Solution: **Docker**
- Very nice paper on such a workflow: Peikert and Brandmaier ([2021](#))

Version control with Git

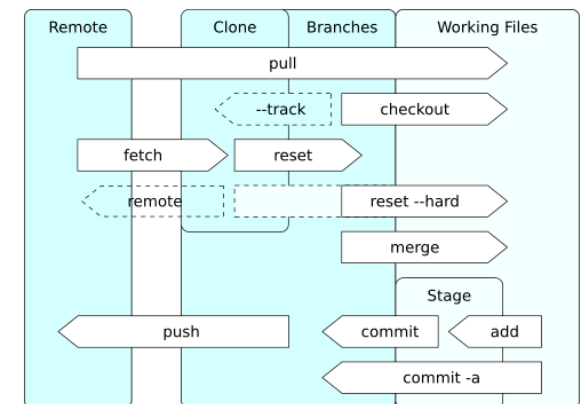
- **De facto standard** version control system
- Can **go back in time** to previous versions
- Can **track changes** between versions
- Can **branch and merge**
- Useful for **collaboration** and **file transfer**
- **Command line** interface (learning curve)
- **Graphical interface** in many editors/IDEs (e.g., RStudio, VS Code)
- **Git** (software) $\neq$ **GitHub/GitLab** (Git repository hosting services)
- Snapshots enable **persistently archiving** (with DOI) on data repositories (e.g., Zenodo)



<https://git-scm.com/downloads/logos>



The Turing Way Community and Scriberia (2024)



[https://commons.wikimedia.org/wiki/](https://commons.wikimedia.org/wiki/File:Git_operations.svg)

File:Git_operations.svg

Automation with Make

- Tool for managing **computational recipes**
- Automating multiple steps into one

Arrabbiata sauce, or sugo all'arrabbiata in Italian, is a spicy sauce for pasta made from garlic, tomatoes, and dried red chili peppers cooked in olive oil.

https://en.wikipedia.org/wiki/Arrabbiata_sauce

Makefile

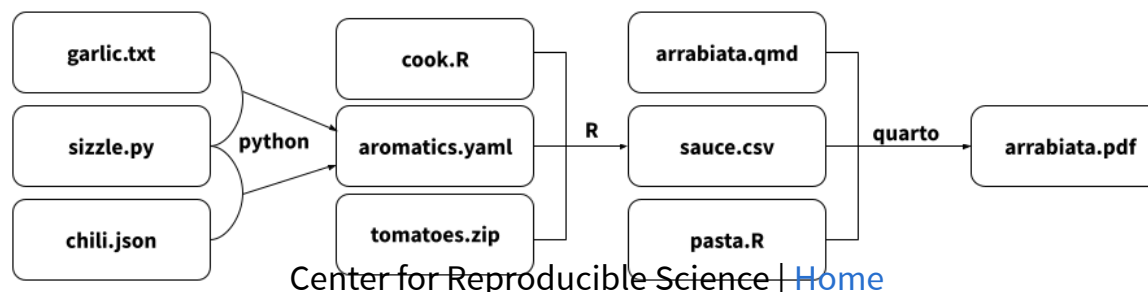
```
arrabiata.pdf: arrabiata.qmd sauce.csv pasta.R
  quarto render arrabiata.qmd

sauce.csv: cook.R tomatoes.zip aromatics.yaml
  R -e "source(cook.R)"

aromatics.yaml: sizzle.py garlic.txt chili.json
  python sizzle.py garlic.txt
  python sizzle.py chili.json
```

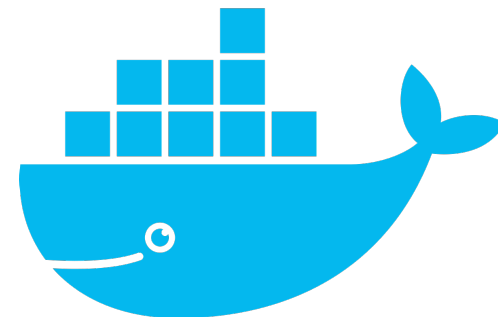
Terminal

```
make arrabiata.pdf # execute recipe
```



Containerization with Docker

- **Software containers** encapsulate computational environment
- Computational environment components:
 - Operating system
 - R and R packages
 - Numerical software libraries (e.g., LAPACK, BLAS)
 - other software (e.g., LaTeX, Java)
- **Docker** is a popular tool for containerization
- **Rocker** is a Docker image for R (<https://rocker-project.org/>)



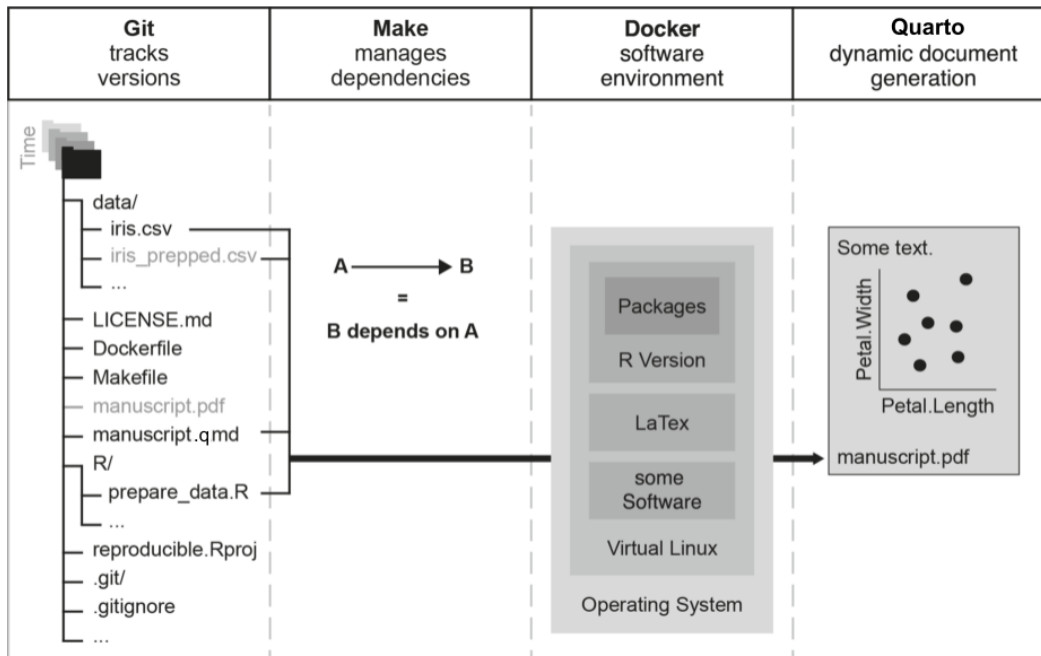
[https://commons.wikimedia.org/wiki/
File:Docker-svgrepo-com.svg](https://commons.wikimedia.org/wiki/File:Docker-svgrepo-com.svg)

Dockerfile

```
# Uses a Rocker container image with R version 4.4
FROM rocker/verse:4.4

# Install R package versions at time when R 4.4 was up-to-date
RUN install2.r ggplot2 dplyr kableExtra
```

Putting it all together



Terminal

```
# clone git repository
git clone git@gitlab.uzh.ch:crsuzh/zurich-names.git

# enter project repository
cd zurich-names

# run automated analyses
make docker-report
```

→ `report.html`

<https://gitlab.uzh.ch/crsuzh/zurich-names>

Modified from Peikert and Brandmaier (2021)

Conclusions

- Dynamic reporting + version control + automation + containerization
 - improve **reproducibility, reuseability, collaboration**
 - lends research more **transparency, credibility**
- There is a **learning curve** 🤔, but ...
 - start simple and learn **step-by-step** 💡
 - tools are **constantly evolving** 🚀
 - invested learning time can **save you time** in the future ⌚
 - skills are **useful in/outside Academia** 💰🧪

The Swiss Reproducibility Network Academy

- **Early career** researchers section of the SwissRN
- Goal 1: Connect young researchers interested in **reproducibility** in **Switzerland**
- Goal 2: Improve reproducibility of research in Switzerland
- How to join?
 - young researchers in **Switzerland** (working language: English)
 - **interest** in improving research and reproducibility
 - **every field** is welcome (diversity is the key!)
- More information: <https://www.swissrn.org/contents/academy/>



References

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The Turing Way Community, and Scriberia. 2024. “Illustrations from the Turing Way: Shared Under CC-BY 4.0 for Reuse.” Zenodo. <https://doi.org/10.5281/ZENODO.3332807>.